

CALL FOR PAPERS SESoS 2018

6th International Workshop on Software Engineering for Systems-of-Systems

<http://sesos2018.systems-of-systems-engineering.org> at ICSE 2018 <http://2018.icse-conferences.org/>

May 27 - 3 June 2018, Gothenburg, Sweden

IMPORTANT DATES

Submission deadline: Feb. 05, 2018
Notification: Mar. 05, 2018
Camera-ready copy: Mar. 19, 2018
Workshop date: May (day), 2018

WORKSHOP ORGANIZERS

Khalil Drira
LAAS-CNRS – Univ. of Toulouse, France
Flavio Oquendo
IRISA – Univ. of South Brittany, France
Paris Avgeriou
Univ. of Groningen, Netherlands
Carlos E. Cuesta
Rey Juan Carlos Univ., Spain

PROGRAM COMMITTEE

Mehmet Aksit, Univ. of Twente, Netherlands
Thais Batista, Federal Univ. of Rio Grande do Norte – UFRN, Brazil
Martin Becker, Fraunhofer IESE, Germany
Isabelle Borne, IRISA – Univ. of South Brittany, France
Ismail Bouassida, Univ. of Sfax, Tunisia
Ivica Crnkovic, Mälardalen Univ., Sweden
Flavia Delicato, Federal Univ. of Rio de Janeiro – UFRJ, Brazil
Laurence Duchien, Univ. Lille – INRIA, France
Cédric Eichler, Univ. Orleans, France
Cristina Gacek, City Univ. London, UK
Amal Gassara, CRNS, Sfax, Tunisia
Nikolaos Georgantas, INRIA Paris, France
Holger Giese, Hasso Plattner Institute at the Univ. of Potsdam, Germany
Itana Gimenes, State Univ. of Maringá, Brazil
Ian Gorton, SEI – CMU, USA
Volker Gruhn, Univ. of Duisburg-Essen, Germany
Wilhelm Hasselbring, Kiel Univ., Germany
Paola Inverardi, Univ. of L'Aquila, Italy
Valérie Issarny, INRIA Paris, France
Rick Kazman, SEI and Univ. of Hawaii, USA
Philippe Kruchten, Univ. of British Columbia, Canada
Patricia Lago, Vrije Univ., Netherlands
Axel Legay, INRIA Rennes, France
Emmanuel Letier, Univ. College London, UK
Grace Lewis, SEI – CMU, USA
Antónia Lopes, Univ. of Lisbon, Portugal
Sam Malek, George Mason Univ., USA
José Carlos Maldonado, Univ. of São Paulo, Brazil
Tomi Mannisto, Univ. of Helsinki, Finland
Raffaella Mirandola, Politecnico Milano, Italy
Henry Muccini, Univ. of L'Aquila, Italy
Jennifer Perez, Technical Univ. of Madrid (UPM), Spain
Claudia Raibulet, Univ. Milano-Bicocca, Italy
Cecilia M.F. Rubira, Univ. of Campinas, Brazil
Robert Nord, SEI – CMU, USA
Salah Sadou, Univ. of South Brittany, France
Richard N. Taylor, Univ. of California, USA
Claudia Maria Lima Werner, Federal Univ. of Rio de Janeiro – UFRJ, Brazil
Danny Weyns, Linnaeus Univ., Sweden
Elisa Yumi Nakagawa, Univ. of São Paulo, Brazil
Uwe Zdun, Univ. of Vienna, Austria
Liming Zhu, NICTA & UNSW, Australia
Andrea Zisman, The Open Univ., UK

SCOPE

Systems-of-Systems (SoS) have become increasingly complex and are frequently used in highly distributed, dynamic, and open environments. SoS have emerged as a new class of evolving software systems, where constituent systems (themselves systems in their own right) work cooperatively in order to fulfill specific missions. SoS are characterized by managerial and operational independence, geographic distribution, evolutionary development, and creation of emergent behavior. These characteristics have therefore raised substantial challenges to software engineering researchers and practitioners. SoS is becoming a hotspot, from both the research and the industrial viewpoints, in particular, related to crucial classes of applications such as safety-critical or mission-critical SoS. Furthermore, SoS are increasingly involved in applications that address societal needs such as environmental monitoring, distributed energy grids management, emergency coordination and crisis management, global traffic control, and smart cities. In this context, the research topic of software engineering for SoS needs to receive more attention and research efforts, considering that the software part of SoS is becoming increasingly larger and more complex. Software engineering theories and tools are needed to adequately design, implement, operate, and even evolve such dynamic, complex software-intensive SoS.. This workshop series is the first that addresses specifically this new perspective. The main goal of SESoS is to provide a forum for software engineering researchers and practitioners to exchange ideas and experiences, analyze research and development issues, discuss promising solutions, and propose inspiring visions for the future.

TOPICS

The workshop will focus on discussing and establishing the foundations for the SoS Software Engineering domain, to pave the way for a more structured community effort. For this, the workshop will focus on the following (and similar) themes:

- SoS Analysis & Architecture
 - Requirements engineering
 - Architectural description
 - Architectural evaluation
 - Architectural evolution
- SoS Model-Based Engineering
 - Modeling and simulation

- Model driven engineering
- Models at runtime
- SoS Construction and Evolution
 - Techniques and technologies
 - Service-orientation
 - Component frameworks
 - Evolutionary development
- SoS Experience
 - Industrial perspectives
 - Successful case studies
- SoS General Issues
 - Characteristics of SoS
 - Quality in SoS
 - SoS development processes
 - Acquisition and project management
 - Verification and validation of SoS
- Future perspective, challenges, and directions of SoS research and/or development

SUBMISSION

Two types of submissions are invited:

- Regular paper (up to eight pages): sound technical contributions from both academia and industry; and
 - Position paper (up to four pages): new ideas, positions, or perspectives of research that are not fully validated.
- All papers submitted to the workshop must be unpublished original work and must not be simultaneously under review or submitted elsewhere. All papers will be judged on the basis of their clarity, relevance, originality, and contribution. All submissions must be in English, formatted according to the ICSE 2018 Format and Submission Guidelines and submitted in either PDF or Postscript format through online upload to the workshop submission website at: <http://sesos2018.systems-of-systems-engineering.org>

PUBLICATION OF PAPERS

The accepted papers will be published in the electronic ICSE Companion Proceedings in both **ACM Digital Library** and **IEEE Digital Library**. SESoS is technically co-sponsored by **ACM SIGSOFT** and **ACM SIGPLAN (*)**.



(*) Previous SESoS editions were technically co-sponsored by **ACM SIGSOFT** and **ACM SIGPLAN**.